2.8.1 Resolving projectile motion

Neglecting air resistance, a projectile has only downward gravitational acceleration. Horizontal velocity stays constant and vertical motion has constant acceleration. Both components share the same elapsed time t.

For launch speed u at angle θ above the horizontal, $u_x = u\cos\theta$ and $u_y = u\sin\theta$. Taking the launch point as the origin and upwards as positive gives $x = u_x t$ and $y = u_y t - \frac{1}{2}gt^2$. At the highest point, vertical velocity is zero, horizontal velocity remains u_x and acceleration still points downwards.

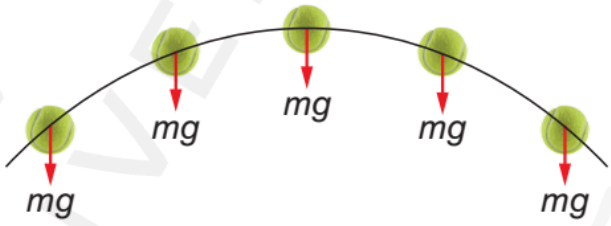


Fig 8.1c *A projectile is acted on only by the gravitational force mg.*

Gravity acts downward at every point, including the highest point. With negligible air resistance, horizontal velocity stays constant.